Name:Shihan Li

Project Title:Abnormal Trading Volume and Short-Term Stock Returns: Evidence from AAPL,JPM,and XOM

1.Hypothesis Pitch

This project investigates whether unusually high trading volume strengthens short-term price continuation after a stock has already experienced a positive daily return.

My hypothesis is:Among positive-return days,stocks with abnormal trading volume will have higher returns over the following five trading days than stocks with normal trading volume.

I focus only on positive-return days because abnormal volume does not always have the same meaning.High volume during a price increase may reflect strong buying interest,new information,or increased investor attention.In contrast,high volume during a price decline may reflect selling pressure or panic.

The economic intuition is that information may not be incorporated into prices immediately.When a price increase is accompanied by unusually high trading activity, the event may attract additional investors and create continued demand over the following days.This idea is related to the high-volume return premium discussed by Gervais,Kaniel,and Mingelgrin(2001).

The hypothesis will be supported if positive-return days with abnormal volume have a higher average subsequent five-day return than positive-return days with normal volume in both the 2024 development period and the 2025 validation period.If the difference is zero or negative in 2025,I will conclude that the hypothesis is not supported out of sample.

2.Data Pipeline

Data Acquisition

The data were obtained from the CRSP Daily Stock File through WRDS.Because WRDS requires authentication,I downloaded the data manually and saved the original CSV file in the project's raw folder.

The sample contains daily observations for three large US companies:Apple Inc. (AAPL),JPMorgan Chase &Co.(JPM),Exxon Mobil Corporation(XOM).

The sample period is from January 2024 to December 2025.The main variables used are ticker,date,daily price,daily return,and daily trading volume.

Data Cleaning

The raw CRSP column names were renamed to simpler names:DlyCalDt became date,DlyPrc became price,DlyRet became ret,DlyVol became volume

The date column was converted to a date format,while price,return,and volume were converted to numeric formats.I then kept only AAPL,JPM,and XOM,restricted the data to the selected two-year period,removed observations missing any required field, and sorted the data by ticker and date.

The two raw CSV files are combined into one dataset.The required columns are renamed,converted to the correct data types,checked for missing values,and sorted by ticker and date.

The cleaned dataset contains 1,755 observations,including 249 warm-up observations from 2023.The formal 2024-2025 research period contains 1,506 observations.

The 2023 observations are used only to calculate historical volume statistics.They are excluded from the hypothesis comparison and backtest results.This allows NAV to be calculated from the beginning of 2024 without losing the first 66 trading days of the research period.

3.Strategy and Backtest

Strategy

The strategy begins by measuring how unusual each stock's daily volume is relative to its own recent history.I use Normalized Abnormal Volume,or NAV:

$$NAV_\{i,t\} = \frac{\{Volume_\{i,t\}-MeanVolume_\{i,t\}\}}{\{StandardDeviationVolume_\{i,t\}\}}$$

The mean and standard deviation are calculated using the previous 66 trading days for the same stock.The current day is excluded from this calculation by shifting the rolling statistics by one day.This prevents the current day's information from affecting its own benchmark.A 66-day window represents approximately one quarter of trading activity and follows the NAV approach used by Wright and Swidler(2023).

The abnormal-volume threshold is the 90th percentile of all valid NAV observations in the 2024 data,which equals 0.94.The cutoff is then frozen and applied unchanged to the 2025 data to avoid the look-ahead bias.

A trading signal occurs when both conditions are satisfied:1.The stock's daily return is positive.2.Its NAV is above the 2024 abnormal-volume cutoff.

The strategy has a clear trading action.If a signal is observed at the end of day(t),the position is entered at the close of the next trading day,(t+1).The position then earns the next five complete daily returns and is sold at the close of (t+6).

The one-day delay makes the rule more realistic because the strategy does not trade using information before it becomes available.New signals for the same stock are ignored while the strategy is waiting to enter or already holding that stock.

Each stock receives a maximum of one-third of the portfolio.Capital that is not invested remains in cash.The strategy does not use leverage or short selling.A round-trip trading cost of 0.10%is deducted from every completed trade.

Equal-Horizon Comparison

Period	Sample Type	Abnormal-Volume Up Days	Average Subsequent 5-Day Return	Normal-Volume Up Days	Average Subsequent 5-Day Return	Abnormal Minus Normal
2024	In-sample	35	0.41%	378	0.46%	-0.05 percentage points
2025	Out-of-sample	42	0.91%	361	0.31%	+0.59 percentage points

The observation counts represent pooled stock-day observations across AAPL,JPM, and XOM,rather than the number of unique calendar days.Therefore,multiple observations may occur on the same trading date.

The main hypothesis test compares:

1.Abnormal-volume group:positive-return days with abnormal trading volume.
2.Normal-volume group:positive-return days without abnormal trading volume.

Both groups use the same subsequent five-day return.The only main difference is whether trading volume was abnormal.

In 2024,the 35 abnormal-volume observations had an average subsequent five-day return of 0.41%,compared with 0.46%for the 378 normal-volume observations.The abnormal-minus-normal difference was -0.05 percentage points.

In 2025,the 42 abnormal-volume observations had an average subsequent five-day return of 0.91%,compared with 0.31%for the 361 normal-volume observations.The difference was +0.59 percentage points.

Therefore,the hypothesis receives positive support in 2025 but not in 2024.The results do not demonstrate a stable relationship across both periods.

Backtest Per formance

Period	Sample Type	Number of Trades	Cumulative Return	Annualised Sharpe Ratio	Maximum Drawdown
2024	In-sample	25	5.11%	1.19	-3.02%
2025	Out-of-sample	24	7.70%	0.97	-6.32%

Each stock receives a maximum allocation of one-third of the portfolio.When a stock does not have an active position,that part of the portfolio remains in cash.The strategy uses no leverage.

A transaction cost of 0.10%is deducted for each completed trade.

In 2024,the strategy completed 25 trades and produced a cumulative return of 5.11%, a Sharpe ratio of 1.19,and a maximum drawdown of -3.02%.

In 2025,the strategy completed 24 trades and produced a cumulative return of 7.70%, a Sharpe ratio of 0.97,and a maximum drawdown of-6.32%.

The positive backtest returns do not by themselves prove the hypothesis.The backtest asks whether the complete trading rule made money,whereas the equal-horizon comparison asks whether abnormal-volume up days performed better than normal-volume up days.

These results suggest that the abnormal-volume condition may help identify stronger positive-return events.Nevertheless,the result is based on only three stocks and two years of data,so it should be treated as preliminary evidence rather than a reliable production strategy.

4.Methodology Defense

Choice of Stocks

There are four reasons why I select these three companies'stocks.

First,they are large and actively traded US companies.Their high liquidity provides reliable daily price and volume data and reduces problems caused by infrequent trading.

Second,the three companies operate in different industries:technology,banking,and energy.This helps prevent the results from depending entirely on the trading patterns of one industry.

Third,all three stocks have complete daily data throughout 2024 and 2025,which allows the same method to be applied consistently across the development and validation periods.

Finally,using three stocks keeps the dataset manageable and makes the data pipeline and backtest easier to understand and explain.The purpose of this project is to demonstrate the research process clearly rather than to build a large production portfolio.

However,three stocks are not representative of the entire US equity market.Therefore, the results should be interpreted as evidence from a small case study.Testing more stocks,including smaller companies and delisted firms,would be an important extension of the research.

Development and Validation Periods

The 2024 data are used to calculate the abnormal-volume cutoff and examine the original relationship.The 2025 data are then evaluated using the same cutoff and trading rules.

A chronological split is appropriate because a real trading strategy must use past information to make later decisions.This allows the project to test whether a rule identified using past data continues to work on later data.Recalculating the cutoff using all 2025 observations would introduce look-ahead bias because the full-year percentile would not have been known at the beginning of 2025.

Sixty-Six-Day Volume Window

A66-trading-day window represents approximately one quarter of trading activity.It provides enough observations to estimate each stock's recent average volume and standard deviation while still allowing the measure to respond to changing market conditions.

NAVMeasure

Raw trading volume cannot be compared fairly across AAPL,JPM,and XOM because the companies normally trade different numbers of shares.NAV expresses each stock's volume relative to its own history and measures the distance from its recent mean in standard-deviation units.

Abnormal-Volume Cutoff

The 90th percentile identifies the highest 10%ofNAV observations in the development period.This makes abnormal events relatively unusual while retaining enough observations for a small project.

The numerical cutoff of 1.06 is obtained only from 2024 data.I do not recalculate it using 2025 data because doing so would allow validation-period information to influence the strategy.

Positive-Return Condition

The study is restricted to positive-return days because the interpretation of high volume depends on price direction.High volume with a price increase may indicate buying pressure,while high volume with a price decline may indicate selling pressure. Separating the directions gives the hypothesis a clearer economic meaning.

Five-Day Holding Period

Five trading days represent approximately one trading week.This is short enough to test short-term continuation but long enough to allow a delayed market response.It also produces more independent observations than a much longer holding period.

The five-day period was used as one simple research horizon rather than selected from many alternatives based on the best backtest result.

Control Group

Positive-return days with normal volume are the main comparison group.They have the same price direction and future measurement period as the signal group.This controls for the possibility that stock prices tend to continue rising after any positive day,even when volume is not unusual.

Execution and Transaction Costs

The strategy waits until the following day's close before entering.This prevents it from assuming that a trade could be placed at the same closing price used to identify the signal.

The 0.10%round-trip cost is a simple estimate covering commissions,bid-ask spread, and slippage for highly liquid US stocks.Actual costs could be higher during volatile market conditions,so this assumption remains a limitation.

5.Pitfall Checklist

Look-ahead Bias

The rolling mean and standard deviation use only the previous 66 trading days because they are shifted by one day.The abnormal-volume cutoff is estimated only with 2024 observations and is frozen before it is applied to 2025.

A signal is observed at the end of day(t),but the trade is not entered until the close of day(t+1).Therefore,the backtest does not assume that the strategy can trade before the signal information is available.

Survivorship Bias

The study uses three companies that remained publicly traded throughout the entire sample.Companies that failed,merged,or were delisted are not included.This may make the results more positive than those from a market-wide historical sample.

Over fitting/Data-Snooping

The main parameters were selected using economic reasoning and prior research rather than by testing many combinations and reporting only the best one.The 66-day window follows the NAV literature,while the top 10%cutoff preserves enough observations for this small sample.

However,I inspected the 2025 results while improving the project.Therefore,2025 is a time-based validation period but is not a perfectly untouched test set.A genuinely new period,such as 2026,would provide a stronger final out-of-sample test.

Transaction Costs and Slippage

CRSP daily returns do not include commissions,bid-ask spreads,slippage,or market impact.The backtest therefore deducts a fixed 0.10%round-trip cost for every completed trade.The cost is recorded when the position begins,but it represents the combined estimated cost of both entering and exiting the trade.

This assumption is reasonable as a simple classroom estimate because AAPL,JPM, and XOM are highly liquid stocks.However,actual costs are not constant. Abnormal-volume events may occur around earnings announcements or unexpected news,when volatility and slippage can be higher.The model also assumes that the strategy can trade at the next day's closing price,which may not be achieved exactly in practice.Therefore,the reported backtest returns may overestimate actual investable performance.

Out-of-Sample Validation

The analysis uses a chronological split.The 2024 data is the development sample, while 2025 is the time-based validation sample.September-December 2023 is used only as warm-up history and is not included in either set of formal results.

The abnormal-volume cutoff of 0.94 is estimated from 2024 and then applied unchanged to 2025.Recalculating the cutoff from the complete 2025 distribution would allow future 2025 information to influence earlier 2025 signals and would prevent a fair test of whether the 2024 rule remained useful.

The 2025 abnormal-volume group outperformed the normal-volume group by 0.59 percentage points,providing positive validation evidence.However,the corresponding 2024 difference was-0.05 percentage points.Therefore,the relationship was not consistent across both periods.In addition,the validation period contains only one year and was viewed during earlier project development,so the findings should be treated as preliminary rather than definitive out-of-sample proof.

6.Efficiency Touchpoint

The cleaned dataset contains 1,755 observations,including 249 warm-up observations from 2023.The formal 2024-2025 research period contains 1,506 observations for three stocks.Most data-cleaning operations,including converting data types,checking missing values,and calculating new columns,process each observation once and have approximately O(n)time complexity.Sorting the data by ticker and date has approximately O(nlog n)complexity.The 66-day rolling mean and standard deviation are calculated with pandas groupby()and rolling()operations,which are more efficient than manually recalculating the previous 66 observations for every row.

The main row-by-row loop in add_strategy_returns checks whether each stock is waiting to be purchased,currently being held,or remaining in cash.Each observation is visited once,so the loop has approximately O(n)time complexity.

7.Risk Touchpoint

The strategy uses no leverage,allocates no more than one-third of the portfolio to each stock,and keeps unused allocations in cash.However,holding only AAPL,JPM, and XOM creates concentration risk,as company-specific or market-wide events may affect the portfolio substantially.

Abnormal volume may reflect news,profit-taking,investor disagreement,or temporary attention rather than continued buying pressure.The different 2024 and 2025 results indicate that the signal is not stable over time.The strategy also faces overnight price gaps and may continue losing because its fixed five-day holding period has no stop-loss or early exit.Finally,actual returns may be lower than the backtest because trades may not occur at the assumed closing price,particularly when spreads and slippage increase during volatile abnormal-volume events.

8.Reflection

My original idea was that unusually high trading volume would generally be followed by a stock-price increase.During the project,I learned that this hypothesis was too broad because high volume can have different meanings when the stock price is rising or falling.High volume during a price decline may represent selling pressure rather than positive investor attention.I therefore refined the question to examine whether abnormal volume on positive-return days is followed by stronger subsequent returns.

I also learned that the choice of comparison group is important.A direct comparison with buy-and-hold would not provide a fair test because buy-and-hold remains invested for much longer than the five-day strategy.The final design instead compares abnormal-volume up days with normal-volume up days using the same future five-day return for both groups.This controls the investment horizon and makes the comparison more closely connected to the hypothesis.

My original backtest plan was to compare the strategy's annual return with the annual return from buying and holding the three stocks for the whole year.However,I later realized that this would not provide a fair test of the hypothesis because the two approaches have very different market exposure.Buy and hold remains fully invested throughout the year,while the abnormal-volume strategy holds a stock for only five days after a signal and otherwise keeps the unused allocation in cash.In a generally rising market,buy-and-hold may earn a higher cumulative return simply because it is exposed to the market for more days.Therefore,buy and hold comparison cannot restrict the difference only in signal quality,but combine the difference in holding time.For this reason,I do not use buy and hold comparison in backtest.

However,I use the equal-horizon comparison in hypothesis testing.
Abnormal-volume up days and normal-volume up days are both evaluated using the same subsequent five-day return.This comparison controls the holding period and more directly tests whether abnormal volume provides additional predictive information.

Adding the 2023 warm-up period was another important improvement.Without warm-up data,the first 66 observations of 2024 could not receive an NAV value.The additional historical data allows the 66-day rolling mean and standard deviation to be calculated from the beginning of the formal research period.The 2023 observations provide only historical information and are not included in the 2024-2025 results.The fact that the results changed after adding warm-up data demonstrates that data preparation can materially affect an empirical conclusion.

The final evidence is mixed.In 2024,35 abnormal-volume up days had an average subsequent five-day return of 0.41%,compared with 0.46%for 378 normal-volume up days.The abnormal-minus-normal difference was -0.05 percentage points.
Because the difference is very small and no formal significance test is conducted,this result is best interpreted as no clear abnormal-volume advantage in 2024 rather than evidence that abnormal volume caused prices to fall.

In 2025,42 abnormal-volume up days produced an average subsequent five-day return of 0.91%,compared with 0.31%for 361 normal-volume up days.The positive difference of 0.59 percentage points supports the hypothesis during the validation period.However,because the abnormal-volume group did not outperform the normal-volume group in both periods,the original hypothesis is not consistently supported.

The trading strategy was profitable in both years after the assumed transaction costs. It returned 5.11%in 2024 and 7.70%in 2025.However,these results answer a different question from the equal-horizon comparison.The comparison tests whether abnormal-volume up days outperform normal-volume up days,while the backtest measures the performance of a complete trading rule that sometimes remains in cash and prevents overlapping positions.Therefore,positive strategy returns do not prove that the hypothesis is universally correct.

The risk results provide an additional lesson.Although the 2025 cumulative return was higher,its Sharpe ratio was lower and its maximum drawdown was larger than in 2024.This shows that evaluating a strategy only by cumulative return can be misleading.Risk-adjusted performance and downside losses must also be considered.

In conclusion,the project suggests that abnormal volume on an up day may result in additionally positive return,but it is not a stable predictor in this sample.The evidence applies only to AAPL,JPM,and XOM during 2024-2025 and does not prove causality or statistical significance.Future research should include more stocks, more market environments,and longer period of time.In addition,the regression could be used to test the correlation and statistical significance between the stock return and multiple factors which could affect the stock performance.

References

- Center for Research in Security Prices(CRSP)(n.d.)CRSP Daily Stock File [online database].Available via:Wharton Research Data Services at: https://wrds-www.wharton.upenn.edu/(Accessed:23 October 2026).
- Gervais,S.,Kaniel,R.and Mingelgrin,D.H.(2001)'The high-volume return premium',The Journal of Finance,56(3),pp.877-919.Available at: https://doi.org/10.1111/0022-1082.00349(Accessed:23 October 2026).
- Wright,C.and Swidler,S.(2023)'Abnormal trading volume,news and market efficiency: Evidence from the Jamaica Stock Exchange',Research in International Business and Finance,64,article 101804.Available at: https://doi.org/10.1016/j.ribaf.2022.101804(Accessed:23 October 2026).